

SHUBHAM MATHUR

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WORK EXPERIENCE

Staff Software Engineer CivilGrid

Apr 2026 - Present
Remote, India

Senior Software Engineer Vimeo

Jan 2024 - Jan 2026
Remote, India

- **Trust & Safety Overhaul (User Lifecycle):** Sole technical owner, authored tech spec and re-architected a broken deactivation pipeline where **195K+ zombie accounts** could still log in (security & GDPR risk). Designed **single source of truth** data model with **DDD** and **RabbitMQ-based async processing**, scaling bulk capacity from 4K to **50K+ accounts**. Reduced deactivation time from days to under **2 minutes**, cleaned **1M+ orphaned records**, and built **Grafana/PagerDuty** observability.
- **Legacy Modernization (Identity):** Designed and implemented “Anumati,” a **Golang** authentication microservice with **REST and gRPC/Protobuf** APIs, decoupling auth from a legacy PHP monolith. Reverse-engineered 15 years of undocumented logic across **40+ integration points** to map the **Strangler Pattern** migration and implemented core auth infrastructure.
- **Regulatory Compliance (Age Assurance):** Authored tech spec and built Vimeo’s age verification platform across **5 platforms** (Web, iOS, Android, tvOS, Vision Pro) for **4.5M+ users** under the **UK Online Safety Act** and EU DSA, mitigating fines of up to **10% of global revenue**. Engineered a **Zero-PII architecture**. Built capability-based **exemption workflows** to bypass verification for premium users, safeguarding revenue streams.
- **Security & Auth Hardening:** Identified a critical **MFA bypass vulnerability** and authored the design specification for a Backup Recovery Code system. Hardened auth flows against **XSS and Open Redirect** vectors. Established source-of-truth **technical specifications** for Compliance and Auth domains, setting engineering standards for the Identity team.

Engineer 2 McKinsey & Company

Nov 2021 - Jan 2024
Bengaluru, India

- Architected **GenAI-powered backend services** for enterprise banking IVR using **NestJS**, designing secure **OpenAI integration layers** with **PII redaction (NER)**, **streaming responses**, and **stateful conversation orchestration** (Redis). Implemented **prompt engineering**, **intent detection**, and **hallucination guardrails** for real-time voice interactions.
- Built **event-driven medication adherence platform** using **Azure Functions, Azure Service Bus, and Twilio**, implementing fault-tolerant message routing and scheduling flows with **HIPAA-compliant** data handling. Engineered automated pharmacy refill triggers and multi-timezone notification routing for elderly care at scale.
- Optimized **NestJS microservices** serving **500K+ monthly users**, reducing API response times by **50%** through caching and index optimization. Led architecture reviews, mentored 4 engineers, established API design standards, and drove CI/CD adoption across teams.

Engineer 1 McKinsey & Company

Jul 2018 - Nov 2021
Bengaluru, India

- Designed **microservices architecture** for Malaysian banking self-service platform serving **20M+ customers**, decomposing legacy SOAP integrations into **REST-based services** and increasing relationship manager productivity by **60%**.
- Developed **insurance recommendation engine** for **1.3M users** using React and Node.js, reducing manual underwriting workflows by **90%**, and built telecom customer platforms for **10M users** with strengthened session security.

EDUCATION

Vellore Institute of Technology | **Bachelor of Technology**, Information Technology

Vellore, India | 2018

SKILLS

Open Source (PRs merged): [Apple/password-manager-resources](#), [Wise/PipelineWise](#), [Reactive Resume](#)

Languages: JavaScript, TypeScript, SQL; **Familiar:** PHP, Go (Golang), Python, Java

Backend: Node.js, Express.js, NestJS; **Familiar:** Spring Boot, FastAPI

Frontend: React, Next.js, Tailwind, Web Security (CSP/XSS), Jest

Databases & Data Warehouse: PostgreSQL, MySQL, Redis, MongoDB; **Familiar:** Snowflake

Queues: Azure Service Bus, RabbitMQ; **Familiar:** Kafka

Cloud & DevOps: Docker, GitHub Actions, CircleCI; **Familiar:** AWS, Azure, GCP, Kubernetes, Grafana

AI/GenAI: Conversational AI, Agentic Workflows, RAG, Function Calling, MCP Servers, Vector DBs & Embeddings